Sustainable Nutrition Market Intelligence

A Tableau BI Case Study on Danone

Lucas Le

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[Github · Linkedin]

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1. Executive Summary

This business intelligence addresses Danone's strategic advantages to identify optimal markets for sustainable nutritional product expansion, aligned with SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption and Production). Analyzing 35 countries across 2016-2020 using 33 indicators from World Bank and FAO databases, the dashboard reveals critical opportunities where malnutrition challenges combine with dairy consumption gaps and resource sustainability considerations. Tanzania, Senegal, and Bangladesh emerge as top-priority markets based on the Sustainable Nutrition Opportunity Index, demonstrating serious dairy protein deficits substantially below global averages. The Strategic Positioning Matrix divides markets into four quadrants including sustainability-focused partnerships for high-risk contexts, scalable commercial products for stable markets, premium sustainability offerings for resource-dependent economies, and innovation testing for developed markets. Recommendations emphasize affordable dairy products targeting children under five, smallholder farmer partnerships, and distribution innovations maintaining environmental stewardship aligned with SDG 12 principles.

2. Introduction & Business Context

Danone, guided by "One Planet. One Health" mission, operates at the intersection of commercial success and social purpose, serving one billion consumers globally whilst maintaining B Corporation certification (Danone, 2024). The corporation faces a strategic imperative to expand nutritional products into emerging markets characterised by malnutrition. Organizations must navigate this expansion sustainably given mounting environmental pressures facing the global dairy industry. The urgency is increasing, globally, 150.2 million children under five suffer from stunting and 733 million people experience hunger, with economic productivity losses exceeding $1 trillion annually (WHO, World Bank, & UNICEF, 2025; World Bank, 2025). However, traditional expansion approaches risk exacerbating environmental challenges. The dairy industry faces stricted ESG supervision, with agricultural emissions projected to comprise 70% of allowable greenhouse gas emissions by 2050 (FAIRR, 2024). Industry executives cite sustainability as their leading priority, driven by consumer preferences, shareholder pressure, and regulatory action including California's mission for 40% methane reduction by 2030 (McKinsey, 2024). This BI solution addresses: "How can Danone systematically identify markets for nutritional dairy expansion that maximise social impact whilst ensuring resource sustainability and commercial viability?" The analysis explicitly integrates SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption), operationalizing Danone's "Danone Impact Journey" sustainability roadmap.

3. Dataset & Data Preparation

3.1 Dataset Overview and Structure

The analysis integrates World Bank Sustainable Development Goals Database (29 indicators) and FAO Food Balances Statistics (4 dairy indicators), supplemented with GDP metrics. The consolidated dataset comprises 180 country-year observations (35 countries × 5 years, 2016-2020) across 36 variables including nutrition security indicators (undernourishment, stunting, wasting), dairy consumption patterns, agricultural capacity (cereal yield, value added per worker), and resource sustainability metrics (natural resource rents, adjusted net savings).

Geographic coverage spans nine regions ensuring representation of both severe malnutrition contexts and developed market benchmarks: Europe (6 countries), Sub-Saharan Africa (6), Southeast Asia (5), Middle East & North Africa (5), Latin America (4), East Asia (3), South Asia (3), North America (2), and Oceania (1).

3.2 Data Cleaning and Transformation

The main challenge was reconciling different database structures. World Bank data arrived in wide format requiring pivoting to long format (World Bank, 2022). FAO data needed the opposite, transposing dairy metrics into separate columns. There are seven country names required harmonisation ensuring clean merging.

Missing values appeared as ".." in World Bank data, converted to blanks for proper null handling. To deliberately avoiding imputation; guessing missing nutrition data for strategic decisions seemed risky.

Rather than pre-merging datasets, the analyst structured data for Tableau's relationship model with three join conditions: Country Name, Country Code, and Year. Outer join logic retained countries with incomplete coverage, preserving 35 countries rather than dropping to approximately 25.

3.3 Quality Validation and Key Assumptions

Validation confirmed no duplicate country-year pairs and consistent ISO3 country codes. Outliers were validated and retained—these extreme cases require understanding.

Critical assumptions: 2016-2020 patterns remain relevant for 2025 planning despite COVID-19 impact. Missing nutrition data isn't random; countries with weakest infrastructure often face worst malnutrition. Country-level averages prioritise initial screening, though priority markets need subnational analysis later.

Data completeness varies: SDG 12 (96%), FAO dairy metrics (92%), but SDG 2 nutrition only 40% due to biennial surveys. Overall 55.8% completeness serves as strategic screening rather than operational planning.

4. Dashboard Design & Functionality

4.1 User Journey and Dashboard Navigation

The dashboard serves two interconnected audiences with distinct analytical needs. Danone executives require rapid strategic overview enabling portfolio-level decisions, whilst market entry strategy teams need granular analytical depth supporting business case development.

The six-page dashboards accessible via bottom navigation, guiding users from high-level strategic overview. A persistent year filter enables temporal analysis across 2016-2020, with an "All" option for aggregated insights. This structure balances executive accessibility with analytical depth, allowing users to drill from the 30,000-foot view to granular country-level metrics.

4.2 Narrative Flow and Storytelling Architecture

The six-page sequence constructs a deliberate analytical narrative. Page 1 (Executive Summary) answers "Where should we look?" through the Sustainable Nutrition Opportunity Index map and Top 10 ranking. Pages 2-4 (SDG 2 Deep Dive) progressively answer "Why these markets?" by unpacking nutrition crisis severity, quantifying dairy intervention opportunities, and assessing supply chain readiness through agricultural infrastructure, establishing both humanitarian imperative and commercial viability.

Page 5 (SDG 12 Sustainability) introduces the critical constraint, revealing resource dependency risks that could undermine long-term market viability. Page 6 (Strategic Connection) synthesises findings through the 2×2 matrix, integrating nutrition opportunity with sustainability risk to answer "How should we prioritise each market?" The four quadrants provide actionable strategic categories.

4.3 Key Performance Indicators and Strategic Metrics

Five calculated fields power market assessment. The Malnutrition Severity Index (0-100 scale) weights undernourishment (30%), stunting (25%), food insecurity (20%), severe wasting (15%), and anemia (10%). The Sustainable Nutrition Opportunity Index synthesises Malnutrition Severity (45%), Dairy Consumption Gap (30%), and inverted Resource Dependency Risk (25%), operationalising Danone's dual commitment to nutrition impact and environmental sustainability. Visualisation choices match analytical intent: filled maps enable geographic pattern recognition, scatter plots reveal paradoxes, and the 2×2 matrix translates complexity into portfolio categories.

5. Descriptive Analytics & Insight Generation

5.1 Malnutrition patterns

Malnutrition hits hardest in Sub-Saharan Africa and South Asia. Congo DRC (16.85), Nigeria (15.16), and Tanzania (14.43) show the most severe conditions. The 2016-2019 trends paint a mixed picture: Bangladesh, Philippines, and Senegal made modest progress, but fragile states went backwards, Congo's undernourishment jumped from 39.9% to 41.7%, Ethiopia from 13.8% to 16.2%, and Nigeria from 9.4% to 14.6%. Political instability clearly undermines nutrition efforts despite SDG commitments.

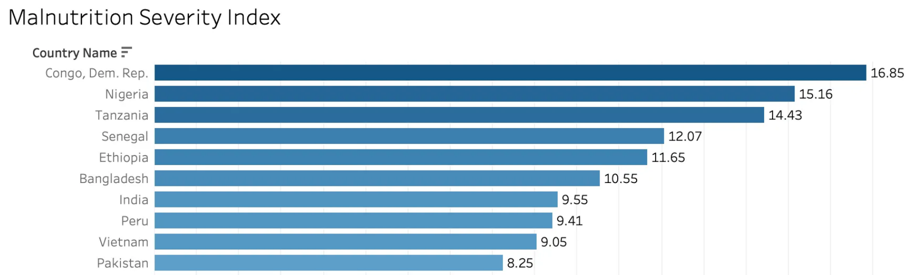

Figure 1 Malnutrition Severity Index Bar Chart

Gender patterns reveal interesting dynamics. Male children face higher stunting risks in most countries (Nigeria +6.27%, Ethiopia +6.15%), but Vietnam and Malaysia show reversed patterns, suggesting cultural factors in food distribution matter more than biology alone. The GDP-stunting scatter reveals a strategic paradox: Pakistan, India, and Colombia consume decent milk (over 100kg/capita/year) yet struggle with high stunting. This isn't a production problem, it's about distribution. Danone's competitive advantage lies in distribution and nutrition innovation rather than infrastructure investment.

5.2 Dairy consumption insights

Dairy consumption reveals dramatic global inequality. Developed markets consume 1,000+ kg/capita annually (UK 1,328, US 1,119), whilst priority markets languish below 150kg such as Nigeria, Tanzania and Senegal. This tenfold gap translates to protein deficiency: Indonesia (1.37g/day), Ethiopia (2.99g) fall severely below the 10.94g average.

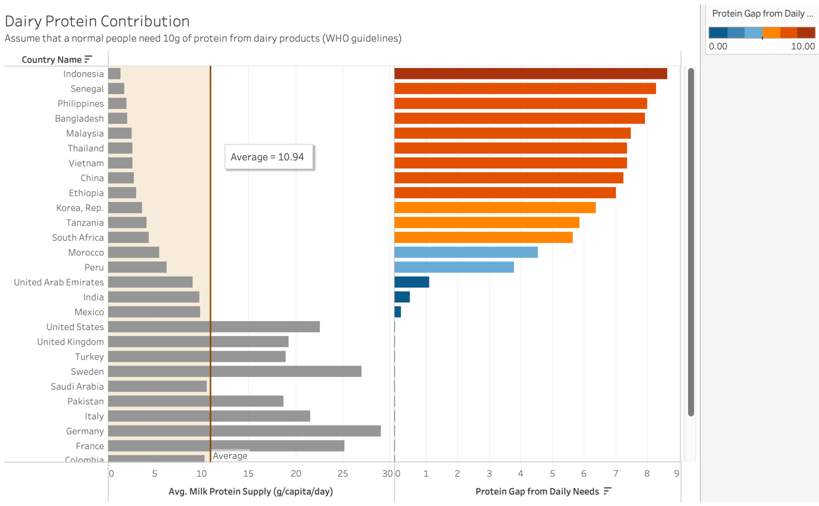

Figure 2 Dairy Protein Contribution Chart

The opportunity sizing matrix identifies "golden quadrant" markets combining high stunting with low consumption: Nigeria, Tanzania, Philippines, Vietnam. India and Pakistan produce substantial milk yet domestic consumption remains suppressed, suggesting distribution bottlenecks. China presents unique opportunity: despite 5% stunting, consumption sits at just 145kg with protein gaps, indicating middle-class growth potential. Mature markets consuming >200kg suggest demand for premium products.

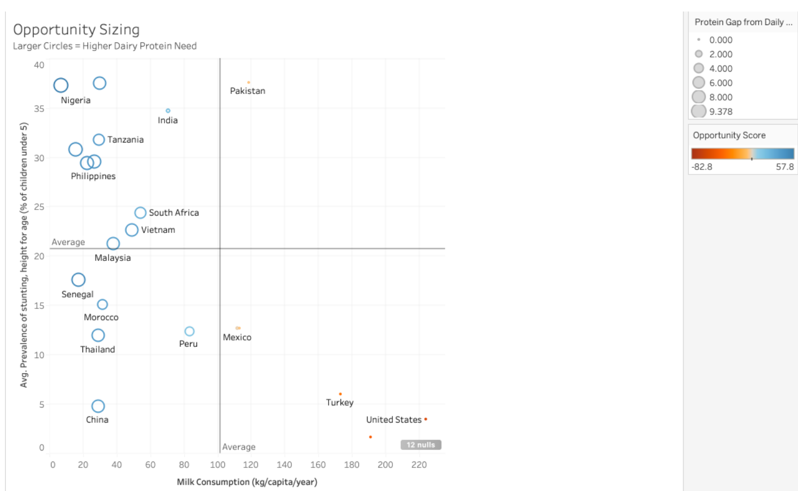

Figure 3 Opportunity Sizing by Dairy Contribution

5.3 Agricultural Paradox & Resource Sustainability

Agricultural infrastructure reveals stark divides. Priority markets combine low cereal yields (<4,000 kg/hectare) and minimal mechanization. The Agricultural Productivity Index quantifies this gap, UAE leads at 30.27 whilst Congo scores 0.77, a forty-fold difference. However, Turkey and Brazil demonstrate weak productivity yet avoid high stunting, revealing the critical paradox.

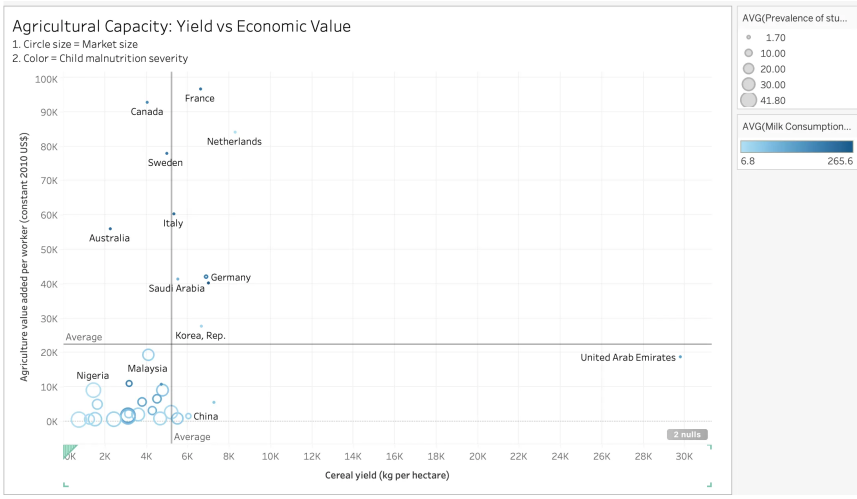

Figure 4 Agricultural Capacity Analysis

The governance mechanism explains this: Brazil's Bolsa Família conditional cash transfer program and National School Feeding Programme ensure nutrition access despite agricultural inefficiencies, demonstrating that targeted social protection systems compensate for weak agricultural productivity (Rasella et al., 2013). Similarly, Turkey's Social Assistance Foundation provides food aid to millions of households (World Bank, 2024). In contrast, India with higher Agricultural Productivity Index (3.11) still experiences elevated stunting, confirming that distribution infrastructure and governance matter more than raw agricultural capacity.

For Danone, this segments strategies: "Infrastructure-ready" markets (India, Pakistan, Vietnam) enable rapid scaling through existing supply chains, whilst "build-from-scratch" markets (Congo, Tanzania, Senegal) require patient capital in smallholder partnerships.

Resource dependency analysis reveals critical risks. Congo DRC scores worst at 81.6, driven by forest dependency and mineral extraction with negative savings. Oil-dependent economies (UAE 70.0, Saudi Arabia 60.2) face "resource curse" vulnerabilities. High-opportunity nutrition markets often overlap with high sustainability risk, whilst Bangladesh demonstrates positive savings indicating diversified economic reinvestment.

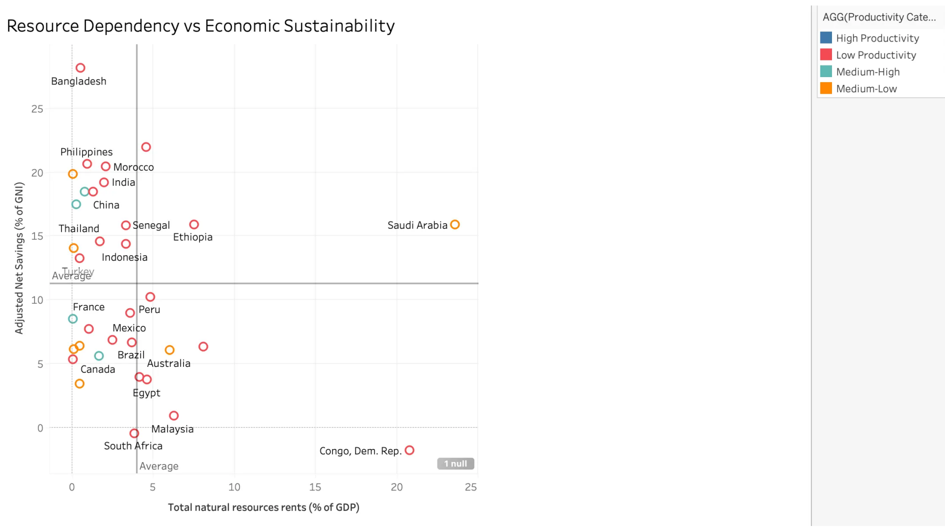

Figure 5 Resource Dependency vs Economic Sustainability Analysis

6. SDG Alignment and Sustainability Analysis

6.1 SDG 2 (Zero Hunger) Performance Assessment

The dashboard directly targeting SDG 2.1 (ending hunger) and SDG 2.2 (ending malnutrition in children under 5). Current performance falls dramatically short: 21.08% average child stunting across sample countries, over four times the WHO acceptable threshold of 5% (WHO, 2022). Temporal analysis shows serious increasing trajectories: Congo's undernourishment increased from 39.9% to 41.7% (2016-2019).

Dairy protein deficits quantify specific intervention opportunities. Countries consuming below 5g/capita/day versus the 10.94g average represent gaps where fortified dairy products directly address SDG 2.2's malnutrition purpose. The dashboard for Malnutrition Severity Index enables tracking a progress toward 2030 benchmarks.

6.2 SDG 12 (Responsible Consumption and Production) Integration

Fifteen countries exceed 10% GDP dependency on non-renewable extraction, with Saudi Arabia (23.0% oil), UAE (13.7% oil), and Congo (12.4% forests) demonstrating unsustainable resource depletion patterns consistent with "resource curse" economics (Sachs & Warner, 2001). The Sustainability Risk Score's adjusted net savings component reveals capital consumption: negative savings in Malaysia, South Africa, and Congo indicate wealth destruction rather than preservation.

High-nutrition-need markets often overlap with high-resource-dependency contexts. The strategic matrix makes this trade-off explicit, Congo, Ethiopia, and Nigeria require sustainability-focused market entry strategies addressing both malnutrition (SDG 2) and resource stewardship (SDG 12) simultaneously.

6.3 Business Intelligence Enabling SDG Monitoring

The dashboard transforms SDG targets from aspirational statements into actionable metrics. Real-time tracking of stunting prevalence, dairy consumption, and resource dependency enables quarterly reporting against 2030 benchmarks. For Danone, SDG alignment is competitive advantage. Investors increasingly search for ESG performance (McKinsey, 2024), consumers demand ethical sourcing, and regulators climate disclosures (U.S. SEC, 2024). The dashboard provides evidence-based storytelling which differentiating Danone from competitors pursuing purely business growth.

7. Strategic Recommendations

7.1 Quadrant-Based Market Entry Strategy

The Strategic Market Positioning Matrix provides a data-driven framework for prioritizing Danone's market expansion across four distinct opportunity categories.

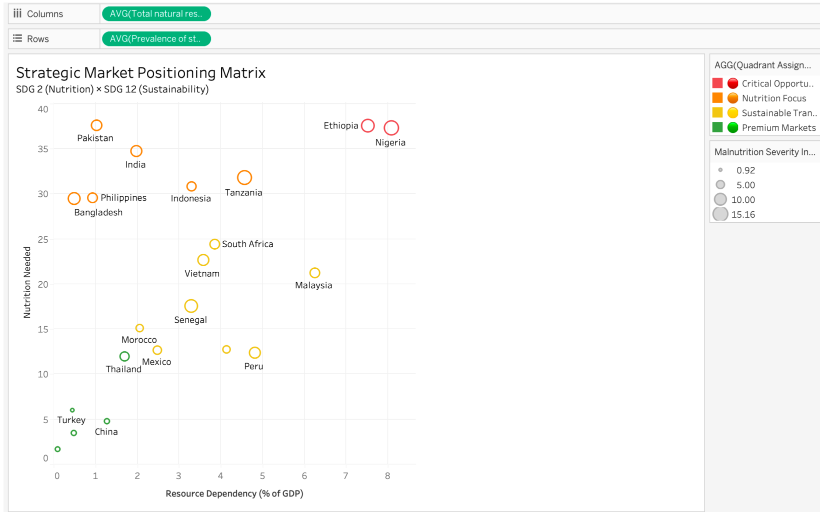

Figure 6 Strategic Market Positioning Matrix

Nutrition Focus Markets (Bangladesh, India, Pakistan, Philippines, Indonesia) emerge as primary expansion targets with severe malnutrition (stunting >25%) alongside relatively stable economic resources (dependency <6% GDP). India and Pakistan show Agricultural Productivity Index scores above 3.0, showing existing supply chain infrastructure can support rapid scaling. Danone should develop affordable fortified dairy products targeting children under five.

Critical Opportunity Markets (Congo DRC, Ethiopia, Nigeria) present several complex challenges requiring sustainable interventions. Resource dependency exceeding 7% GDP combined with malnutrition severity above 11 points demands partnerships with development finance institutions to co-invest in smallholder dairy cooperatives implementing regenerative agriculture practices. Tanzania offers a suitable pilot environment for testing carbon-neutral supply chains before higher-risk expansion.

Sustainable Transition Markets (Saudi Arabia, UAE) offer opportunities for premium sustainability-branded products. Despite low malnutrition rates, oil dependency ranging from 13.7% to 23.0% GDP creates consumer demand for ESG brands.

Premium Markets (China, Thailand, Turkey) enable product innovation testing for functional dairy addressing specific and personalized micronutrient needs rather than basic nutrition problems.

7.2 Implementation & Monitoring

The dashboard facilitates quarterly performance tracking: stunting prevalence changes, dairy protein supply increases, resource dependency risks, and Sustainable Nutrition Opportunity Index progression. Click-through function of dashboard allows rapid assessment against 2030 SDG benchmarks, supporting portfolio adjustments as market conditions evolve. Significant barriers include fragmented dairy supply chains in low-productivity contexts, regulatory complexity, and B Corp recertification requirements (B Lab, 2023). A phased five-year approach to prioritise Nutrition Focus markets (2025-2027) before expanding to Critical Opportunity contexts (2028-2030) will mitigates these challenges while building operational capabilities progressively.

8. Ethical Considerations & Critical Reflection

8.1 Data Limitations and Methodological Constraints

Several methodological choices warrant critical examination. The reliance on country-level aggregates masks significant within-country variation, India's dashboard-reported stunting figures obscure stark urban-rural divides where prevalence may differ substantially from national averages. This aggregation could lead Danone to overlook sub-national markets with acute needs or enter markets where average figures misrepresent accessibility challenges in remote regions.

The 55.8% overall data completeness raises reliability concerns, particularly for SDG 2 nutrition indicators at 40% completeness. Missing data is not random, countries with weakest statistical infrastructure often face most severe malnutrition, creating systematic underestimation of global crisis severity. Additionally, the 2016-2020 timeframe excludes COVID-19's impact, which increased global undernourishment by 118 million people in 2020 alone (FAO et al., 2021).

8.2 Ethical Implications of Commercial Prioritization

The Sustainable Nutrition Opportunity Index inherently prioritises markets offering commercial viability alongside social impact. Congo DRC scores 49.28 despite extreme need (16.85 malnutrition severity) partly due to high sustainability risk deterring investment. While pragmatic for corporate strategy, this raises questions about whether BI-driven frameworks adequately serve "leaving no one behind" principles central to SDG frameworks (United Nations, 2025).

8.3 Transparency and Accountability

Over-reliance on World Bank and FAO data ensures reproducibility but result in data collection biases. Organizations often priories urban populations over marginalised communities. Future analysis should integrate qualitative data from local NGOs to validate findings and ensure interventions address lived experiences rather than statistical abstractions.

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APPENDIX A: Calculated Field Formulas

1. Malnutrition Severity Index (0-100 scale)

Malnutrition Severity Index | Main Data (Danone_BI_Dashboard_Data_CLEAN)

```
// Malnutrition Severity Index (0-100 scale)
(
  IFNULL([Prevalence of undernourishment (% of population)], 0) * 0.30 +
  IFNULL([Prevalence of stunting, height for age (% of children under 5)], 0) * 0.25 +
  IFNULL([Prevalence of severe food insecurity in the population (%)], 0) * 0.20 +
  IFNULL([Prevalence of severe wasting, weight for height (% of children under 5)], 0) * 0.15 +
  IFNULL([Prevalence of anemia among women of reproductive age (% of women ages 15-49)], 0) * 0.10
)
```

Figure 7 Malnutrition Severity Index

Rationale: Aggregates five critical nutrition indicators reflecting WHO malnutrition framework. Undernourishment (30%) receives highest weight as primary SDG 2.1 target. Stunting (25%) reflects irreversible developmental damage. Severe food insecurity (20%) captures multidimensional access issues. Severe wasting (15%) indicates acute malnutrition. Anemia in women (10%) represents hidden hunger and intergenerational transmission.

2. Resource Dependency Risk Score (0-100 scale)

Resource Dependency Risk | Main Data (Danone_BI_Dashboard_Data_CLEAN)

```
// Resource Dependency Risk (0-100 scale)
(
  IFNULL([Total natural resources rents (% of GDP)], 0) * 0.60 +
  (
    100 - IFNULL([Adjusted net savings, excluding particulate emission damage (% of GNI)], 50)
  ) * 0.40
)
```

Figure 8 Resource Dependency Risk Score

Rationale: Resource rents (60%) quantify extraction dependence normalized to 30% GDP threshold where "resource curse" effects become severe. Adjusted net savings (40%, inverted) penalizes capital depletion: negative savings (consuming wealth) scores maximum risk (40 points); ≥20% savings (adequate reinvestment) scores zero; intermediate values scale linearly.

3. Sustainable Nutrition Opportunity Index (0-100 scale)

ainable Nutrition Opportunity Index | Main Data (Danone_BI_Dashboard_Data_CLEAN)

```
// Sustainable Nutrition Opportunity Index (0-100)
// Combines malnutrition need, dairy gap, and resource sustainability

(
  [Malnutrition Severity Index] * 0.45 +
  [Dairy Consumption Gap] * 0.30 +
  (100 - [Resource Dependency Risk]) * 0.25
)
```

Figure 9 Sustainable Nutrition Opportunity Index

Rationale: Composite metric balancing three strategic dimensions. Malnutrition Severity (45%) receives highest weight, reflecting Danone's mission and SDG 2 commitment, ensuring need-based targeting. Dairy Consumption Gap (30%) represents market growth potential. Resource Sustainability (25%, inverted risk) favors diversified economies, ensuring SDG 12 alignment. Higher scores identify markets where nutrition intervention delivers maximum social impact within commercially sustainable contexts.

4. Agricultural Productivity Index (0-100 scale)

Agricultural Productivity Index | Main Data (Danone_BI_Dashboard_Data_CLEAN)

```
// Composite score combining yield and value-added

(
  // Cereal Yield component (50%)
  (IFNULL([Cereal yield (kg per hectare)], 0)
   / { MAX([Cereal yield (kg per hectare)]) }) * 50

  +

  // Ag Value Added component (50%)
  (IFNULL([Agriculture value added per worker (constant 2010 US$)], 0)
   / { MAX([Agriculture value added per worker (constant 2010 US$)]) }) * 50
)

// Score 0-100
// Higher = Better agricultural productivity
```

Figure 10 Agricultural Productivity Index

Rationale: Combines land productivity with labor efficiency to assess agricultural capacity for dairy supply chains. Cereal yield (50%) measures land-use efficiency (kg/hectare), indicating climate suitability and irrigation infrastructure. Agricultural value added per worker (50%) captures mechanization and processing capacity critical for cold storage and distribution networks. Equal weighting reflects that high yield without processing limits scalability, while high mechanization in unsuitable climates faces production constraints. Normalization (0-100) enables cross-country comparison.

5. Sustainability Risk Score (0-100 scale)

Sustainability Risk Score   Main Data (Danone_BI_Dashboard_Data_CLEAN)

```
(
  // Part 1: Resource Rents
  (ZN(AVG([Total natural resources rents (% of GDP)])) / 30) * 60
)
+
(
  // Part 2: Adjusted Net Saving
  IF ZN(AVG([Adjusted net savings, excluding particulate emission damage (% of GNI)])) <= 0
  THEN 40

  ELSEIF ZN(AVG([Adjusted net savings, excluding particulate emission damage (% of GNI)])) >= 20
  THEN 0

  ELSE
    ((20 - ZN(AVG([Adjusted net savings, excluding particulate emission damage (% of GNI)]))) / 20) * 40
  END
)
```

Figure 11 Sustainability Risk Score

Rationale: Identifies countries with unsustainable economic models requiring diversification (SDG 12.2). Resource rents (60%) quantify extraction dependence normalized to 30% GDP where "resource curse" effects become severe. Adjusted net savings (40%, inverted) penalizes capital depletion: negative savings scores maximum risk (40 points); ≥20% savings scores zero. Higher scores indicate vulnerability to commodity shocks and inadequate economic diversification.